败血症存活率数据集分析

来源：
https://archive.ics.uci.edu/dataset/827/sepsis+survival+minimal+clinical+records

s41598-020-73558-3_sepsis_survival_study_cohort.csv (a subset of the primary cohort)

数据集介绍

该数据集包含2011年至2012年间挪威84,811名住院受试者的110,204例入院病例。我们分析的数据集是主数据的一个子集，包含19,051名患者入院的特征。

列信息描述:

age_years: 患者年龄（岁）

sex_0male_1female: 患者性别（0:男性, 1:女性）

episode_number: 既往败血症发作次数

hospital_outcome_1alive_0dead: 住院9天后生存状态（1:存活, 0:死亡）

研究背景与意义

败血症是一种危及生命的器官功能障碍，由感染引起的不受调节的宿主反应导致。早期识别高风险患者对于改善预后至关重要。

核心研究问题

1.年龄与死亡率的关系：不同年龄段的败血症患者死亡率是否存在显著差异?

2.性别差异分析：男性和女性败血症患者的死亡率是否有显著不同?

3.发作次数的影响：既往败血症发作次数是否影响当前住院的生存率?

一. 导入必要package

In [1]:
```python
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from scipy import stats
import warnings
warnings.filterwarnings("ignore")
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
```

二. 读取原始数据

In [2]:
```python
data=pd.read_csv('sepsis_data.csv')
```

三. 数据集清洗

1.数据基本情况:

In [3]:
```python
# 检查数据基本信息
print("数据前5行：")
print(data.head())

print("\n数据基本信息：")
print(data.info())

print("\n描述性统计：")
print(data.describe())
```

```
数据前5行：
   age_years  sex_0male_1female  episode_number  hospital_outcome_1alive_0dead
0          7                  1               1                              1
1         17                  0               2                              1
2         70                  0               1                              1
3         76                  0               1                              1
4          8                  0               1                              1

数据基本信息：
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19051 entries, 0 to 19050
Data columns (total 4 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   age_years                     19051 non-null  int64
 1   sex_0male_1female             19051 non-null  int64
 2   episode_number                19051 non-null  int64
 3   hospital_outcome_1alive_0dead 19051 non-null  int64
dtypes: int64(4)
memory usage: 595.5 KB
None

描述性统计：
          age_years  sex_0male_1female  episode_number  \
count  19051.000000       19051.000000    19051.000000
mean      72.503228           0.448585        1.396252
std       18.614470           0.497363        0.783749
min        0.000000           0.000000        1.000000
25%       65.000000           0.000000        1.000000
50%       77.000000           0.000000        1.000000
75%       85.000000           1.000000        2.000000
max      100.000000           1.000000        5.000000

       hospital_outcome_1alive_0dead
count                   19051.000000
mean                        0.810719
std                         0.391742
min                         0.000000
25%                         1.000000
50%                         1.000000
75%                         1.000000
max                         1.000000
```

2. 检查缺失值

In [4]:
```python
# 检查缺失值
print("缺失值检查：")
print(data.isnull().sum())
```

```
缺失值检查：
age_years                       0
sex_0male_1female               0
episode_number                  0
hospital_outcome_1alive_0dead   0
dtype: int64
```

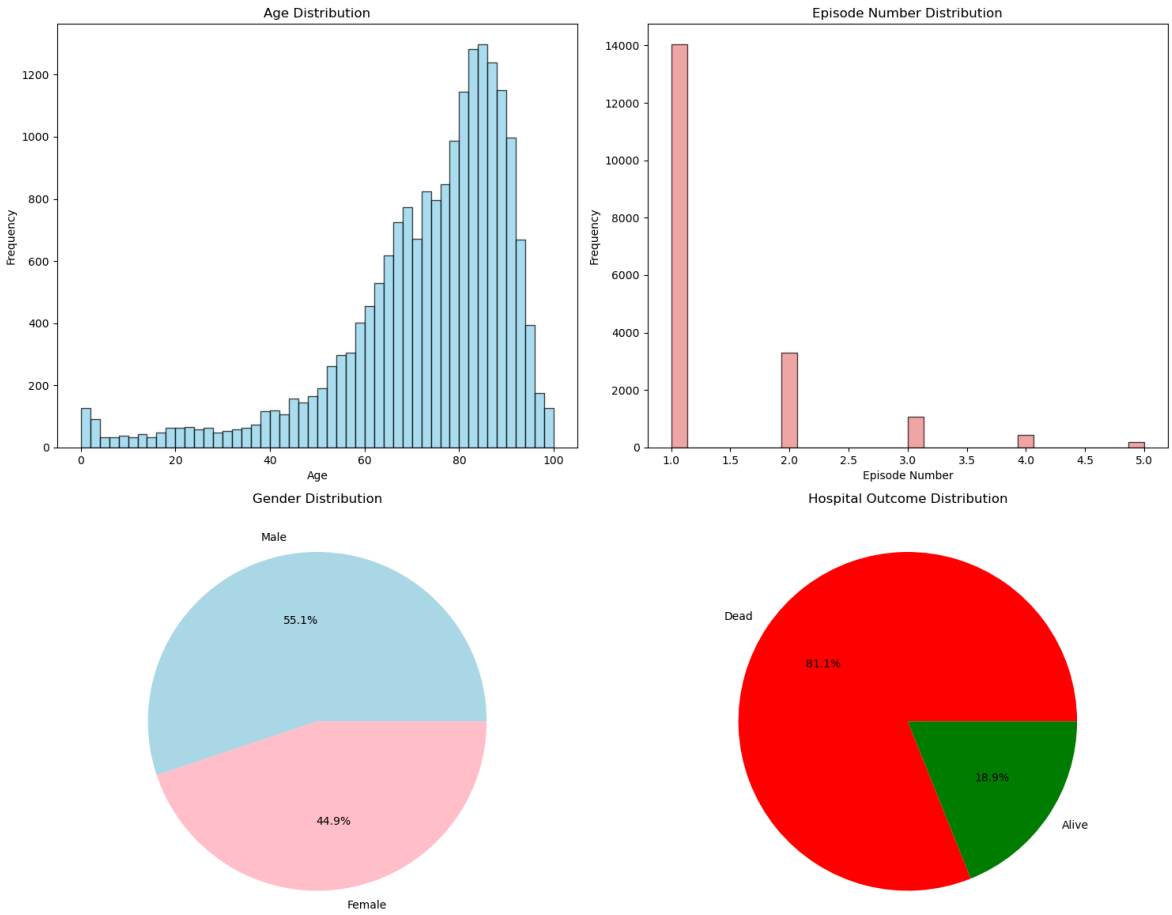
3. 检查数据类型

In [5]:
```
# 检查数据类型
print("\n数据类型：")
print(data.dtypes)
```

```
数据类型：
age_years                        int64
sex_0male_1female                int64
episode_number                   int64
hospital_outcome_1alive_0dead    int64
dtype: object
```

4. 数据质量可视化

In [6]:
```
# 数据质量可视化
fig, axes = plt.subplots(2, 2, figsize=(15, 12))

# 年龄分布
axes[0,0].hist(data['age_years'], bins=50, alpha=0.7, color='skyblue', edgecolor
axes[0,0].set_title('Age Distribution')
axes[0,0].set_xlabel('Age')
axes[0,0].set_ylabel('Frequency')

# 发作次数分布
axes[0,1].hist(data['episode_number'], bins=30, alpha=0.7, color='lightcoral', e
axes[0,1].set_title('Episode Number Distribution')
axes[0,1].set_xlabel('Episode Number')
axes[0,1].set_ylabel('Frequency')

# 性别分布
gender_counts = data['sex_0male_1female'].value_counts()
axes[1,0].pie(gender_counts.values, labels=['Male', 'Female'], autopct='%1.1f%%'
axes[1,0].set_title('Gender Distribution')

# 结局分布
outcome_counts = data['hospital_outcome_1alive_0dead'].value_counts()
axes[1,1].pie(outcome_counts.values, labels=['Dead', 'Alive'], autopct='%1.1f%%'
axes[1,1].set_title('Hospital Outcome Distribution')

plt.tight_layout()
plt.show()
print("无需进行清洗")
```

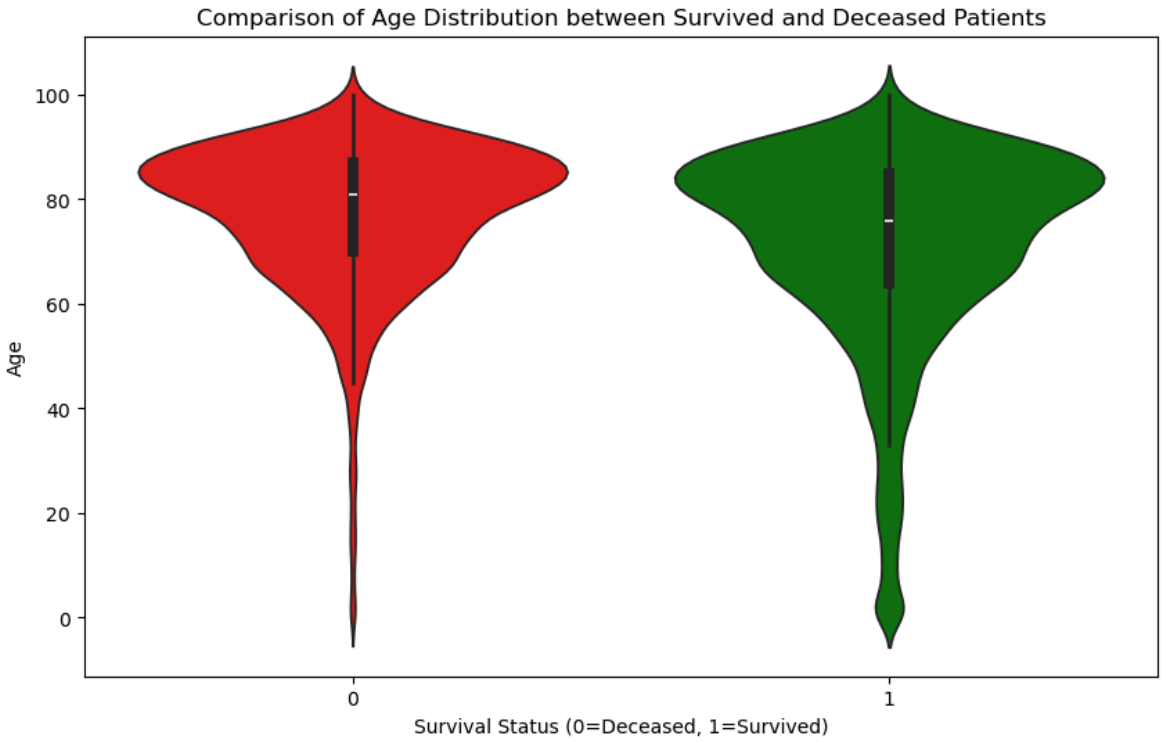


无需进行清洗

四.探索性分析

1.年龄分布在存活与死亡患者中的比较

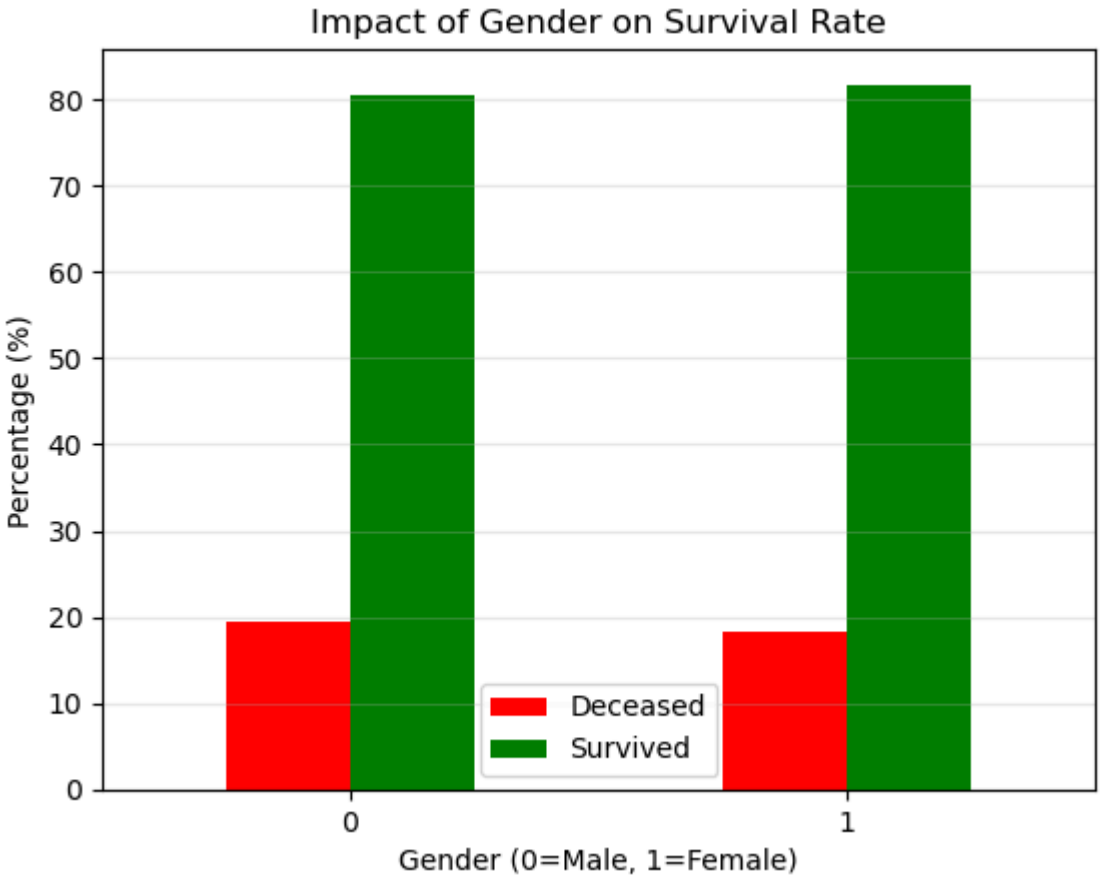
In [7]:
```
# 绘制小提琴图 - 用于比较存活与死亡患者的年龄分布
# 小提琴图结合了箱线图和核密度估计，能同时显示数据的分布形状和关键统计量
plt.figure(figsize=(10, 6))
sns.violinplot(x='hospital_outcome_1alive_0dead', y='age_years', data=data,
               palette={'0': 'red', '1': 'green'})# 红色(0): 死亡患者组，绿色(1)
plt.title('Comparison of Age Distribution between Survived and Deceased Patients
plt.xlabel('Survival Status (0=Deceased, 1=Survived)')
plt.ylabel('Age')
plt.show()

#比较死亡组(红色)和存活组(绿色)的年龄分布差异，小提琴宽度表示患者密度
```



2. 性别对死亡率的影响分析

In [8]:
```
# 创建交叉表
gender_survival = pd.crosstab(data['sex_0male_1female'],
                              data['hospital_outcome_1alive_0dead'],
                              normalize='index') * 100

print("性别与生存率的交叉表（百分比）:")
print(gender_survival)

# 绘制分组条形图
plt.figure(figsize=(8, 6))
gender_survival.plot(kind='bar', color=['red', 'green'])
plt.title('Impact of Gender on Survival Rate')
plt.xlabel('Gender (0=Male, 1=Female)')
plt.ylabel('Percentage (%)')
plt.legend(['Deceased', 'Survived'])
plt.xticks(rotation=0)
plt.grid(axis='y', alpha=0.3)
plt.show()

# 分析不同性别患者的生存率差异，红色条形表示死亡率，绿色表示存活率
```

```
性别与生存率的交叉表（百分比）:
hospital_outcome_1alive_0dead          0          1
sex_0male_1female
0                              19.466921  80.533079
1                              18.265855  81.734145
<Figure size 800x600 with 0 Axes>
```

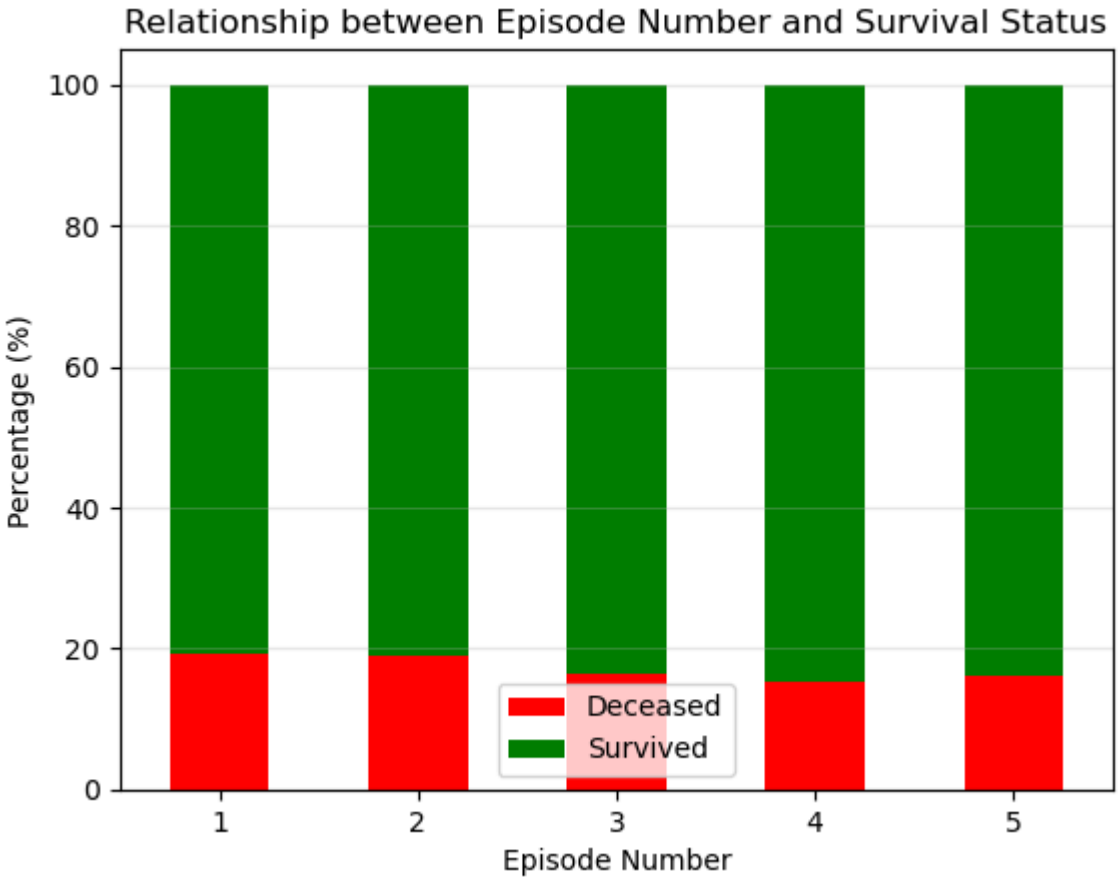


3. 发病次数与死亡率的关系

In [9]:
```
# 使用交叉表分析
cross_tab = pd.crosstab(data['episode_number'],
                        data['hospital_outcome_1alive_0dead'],
                        normalize='index') * 100

print("发病次数与生存状态的交叉表（百分比）:")
print(cross_tab)

# 绘制堆叠条形图
plt.figure(figsize=(10, 6))
cross_tab.plot(kind='bar', stacked=True, color=['red', 'green'])
plt.title('Relationship between Episode Number and Survival Status')
plt.xlabel('Episode Number')
plt.ylabel('Percentage (%)')
plt.legend(['Deceased', 'Survived'])
plt.xticks(rotation=0)
plt.grid(axis='y', alpha=0.3)
plt.show()

#展示不同发病次数患者的生存状态分布，红色部分为死亡率，绿色为存活率
```

```
发病次数与生存状态的交叉表（百分比）:
hospital_outcome_1alive_0dead          0          1
episode_number
1                              19.299245  80.700755
2                              18.855932  81.144068
3                              16.307404  83.692596
4                              15.192744  84.807256
5                              16.243655  83.756345
```

```
<Figure size 1000x600 with 0 Axes>
```



五.分析指标设计

1.多因素综合分析

In [10]:
```
# 多因素逻辑回归 - 控制混杂因素

# 准备数据
X = data[['age_years', 'sex_0male_1female', 'episode_number']]
y = data['hospital_outcome_1alive_0dead']

# 标准化
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 多因素逻辑回归
multi_model = LogisticRegression()
multi_model.fit(X_scaled, y)

# 输出结果
feature_names = ['年龄', '性别', '发病次数']
print("多因素逻辑回归系数 (已标准化):")
for i, (name, coef) in enumerate(zip(feature_names, multi_model.coef_[0])):
    importance = abs(coef)
    print(f"  {name}: {coef:.4f} (重要性: {importance:.4f})")

# 找出最显著的风险因素
most_important = feature_names[np.argmax(abs(multi_model.coef_[0]))]
print(f"最重要的风险因素: {most_important}")
```

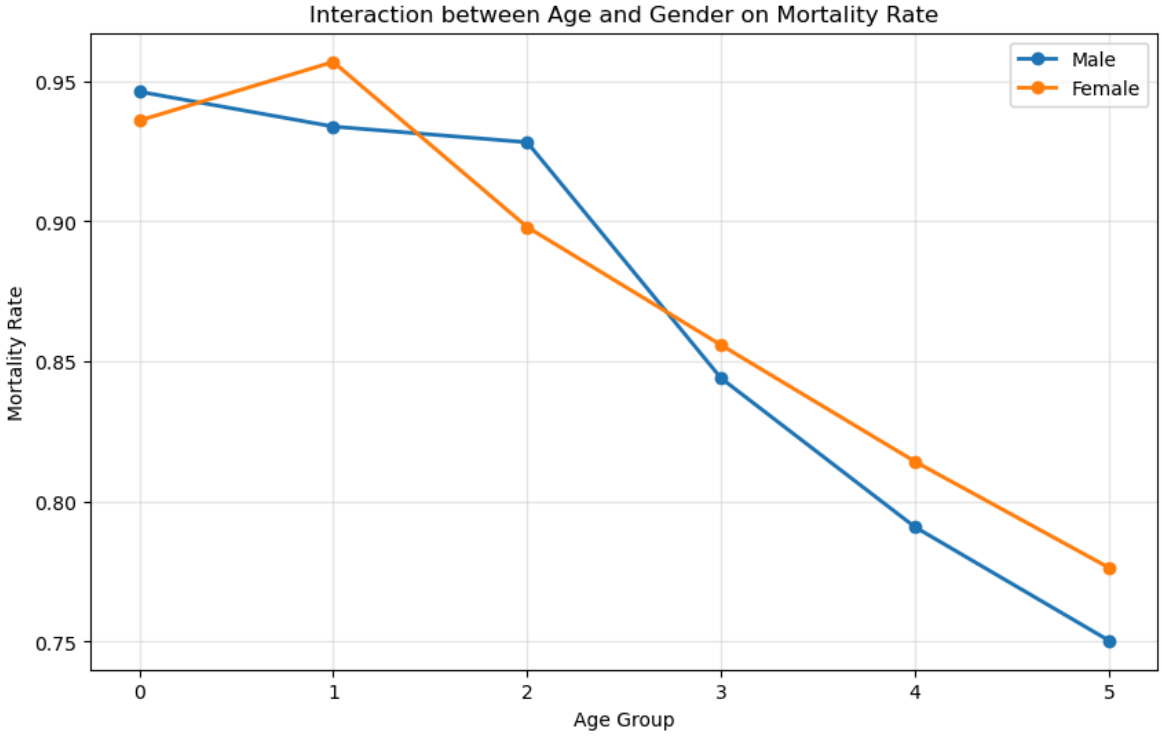
```
# 注释：多因素逻辑回归分析年龄、性别和发病次数对死亡率的影响，系数绝对值越大表示影
```

```
多因素逻辑回归系数 (已标准化):
  年龄: -0.4047 (重要性: 0.4047)
  性别: 0.0693 (重要性: 0.0693)
  发病次数: 0.0391 (重要性: 0.0391)
最重要的风险因素: 年龄
```

2. 交互作用分析

In [11]:

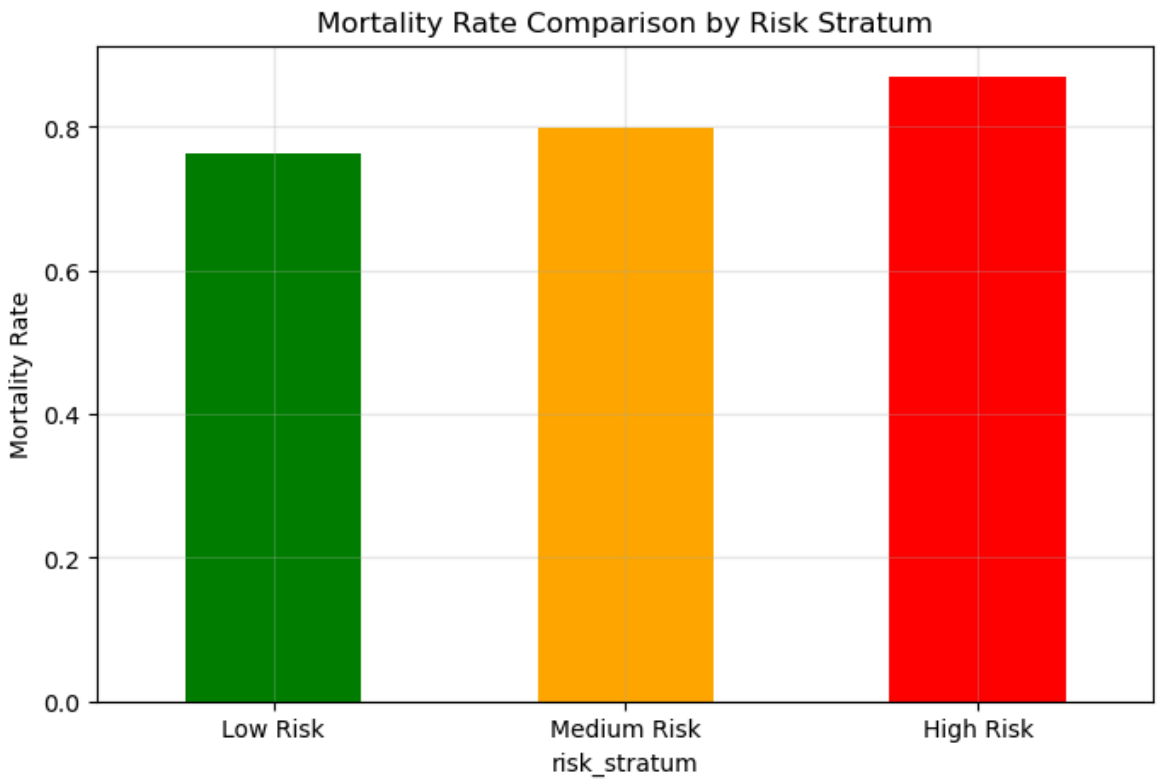
```
# 可视化交互作用
plt.figure(figsize=(10, 6))

# 按性别分组绘制年龄与死亡率关系
gender_labels = {0: 'Male', 1: 'Female'}

for gender in [0, 1]:
    subset = data[data['sex_0male_1female'] == gender]
    if len(subset) > 0:
        # 按年龄分组计算死亡率
        age_bins = pd.cut(subset['age_years'], bins=6)
        age_hospital_outcome_1alive_0dead = subset.groupby(age_bins)['hospital_o

        # 绘制线条
        plt.plot(range(len(age_hospital_outcome_1alive_0dead)), age_hospital_out
                marker='o', label=gender_labels[gender], linewidth=2)

plt.xlabel('Age Group')
plt.ylabel('Mortality Rate')
plt.title('Interaction between Age and Gender on Mortality Rate')
plt.legend()
plt.grid(True, alpha=0.3)
plt.show()
# 注释：分析年龄对死亡率的影响在不同性别间的差异，展示性别与年龄的交互作用
```



3. 风险分层分析

In [12]:
```
# 计算综合风险评分（基于多因素模型）
risk_factors = ['age_years', 'sex_0male_1female', 'episode_number']
X_risk = data[risk_factors]
X_risk_scaled = scaler.transform(X_risk)

risk_scores = multi_model.predict_proba(X_risk_scaled)[:, 1]
data['risk_score'] = risk_scores

# 风险分层
data['risk_stratum'] = pd.qcut(data['risk_score'], q=3,
                               labels=['Low Risk', 'Medium Risk', 'High Risk'])

# 各层死亡率
stratum_hospital_outcome_1alive_0dead = data.groupby('risk_stratum')['hospital_o
print("风险分层结果:")
for stratum, row in stratum_hospital_outcome_1alive_0dead.iterrows():
    print(f"  {stratum}: {row['mean']:.3%} (n={row['count']})")

# 高风险人群特征分析
high_risk = data[data['risk_stratum'] == 'High Risk']
print(f"\n高风险人群特征:")
print(f"  • 平均年龄: {high_risk['age_years'].mean():.1f}岁")
print(f"  • 性别分布: 男性{len(high_risk[high_risk['sex_0male_1female']==0])}人,
print(f"  • 平均发病次数: {high_risk['episode_number'].mean():.2f}次")
print(f"  • 实际死亡率: {high_risk['hospital_outcome_1alive_0dead'].mean():.3%}"

# 可视化风险分层
plt.figure(figsize=(8, 5))
stratum_hospital_outcome_1alive_0dead['mean'].plot(kind='bar', color=['green', '
plt.title('Mortality Rate Comparison by Risk Stratum')
plt.ylabel('Mortality Rate')
plt.xticks(rotation=0)
plt.grid(True, alpha=0.3)
plt.show()
```

```
风险分层结果:
  Low Risk: 76.370% (n=6352.0)
  Medium Risk: 79.915% (n=6373.0)
  High Risk: 86.959% (n=6326.0)

高风险人群特征:
  • 平均年龄: 52.4岁
  • 性别分布: 男性3118人, 女性3208人
  • 平均发病次数: 1.53次
  • 实际死亡率: 86.959%
```



六.结果评估与检验

1.提出假设

编号 零假设 H_0 对应统计方法 H1 年龄与住院结局无关 卡方检验 H2 性别与住院结局无关 卡方检验 H3 发作次数与住院结局无关 卡方检验

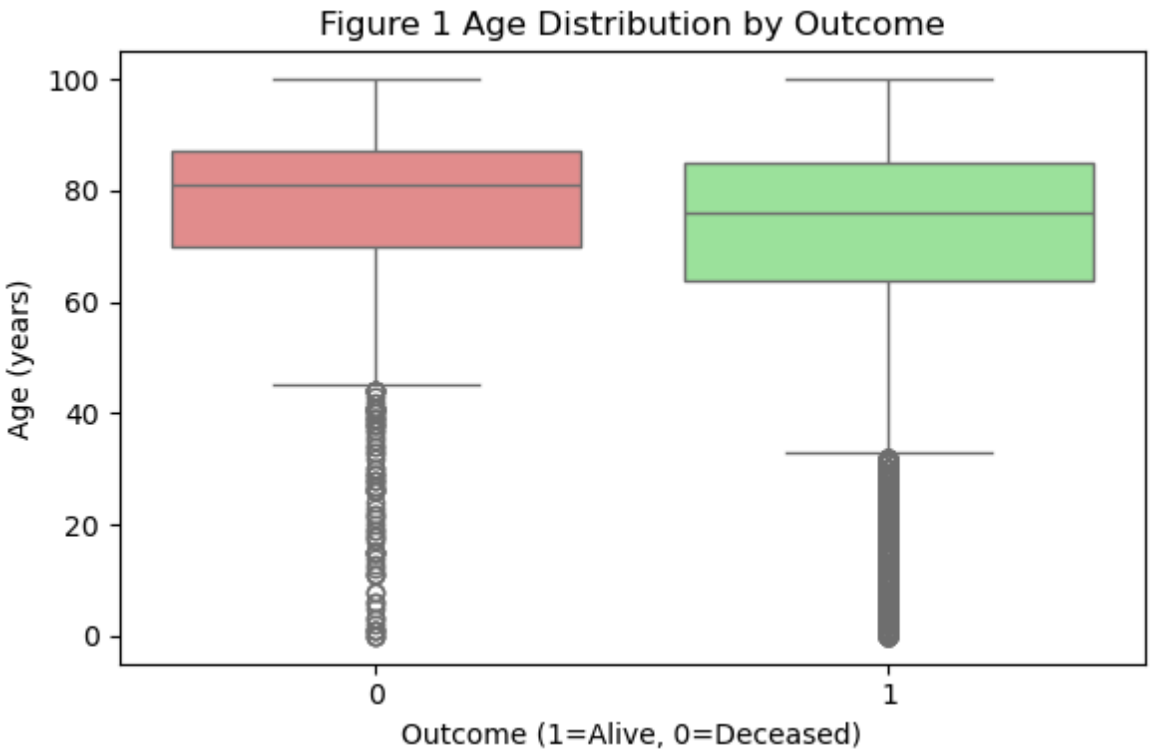
2.卡方检验 + 效应量

In [13]:
```
# 生成年龄分组（用于后续卡方检验）
data['age_group'] = pd.cut(data['age_years'],
                           bins=[0,18,35,60,80,150],
                           labels=['儿童','青年','中年','老年','高龄'])

def chi2_effect(x, y, name_x):
    """卡方检验并计算Cramér's V"""
    tab = pd.crosstab(x, y)
    chi2, p, dof, _ = stats.chi2_contingency(tab)
    n = tab.sum().sum()
    cramers_v = np.sqrt(chi2 / (n * (min(tab.shape) - 1)))
    print(f'{name_x} vs outcome: χ²={chi2:.3f}, p={p:.4f}, Cramér\'s V={cramers_
    return p, cramers_v

# 三组假设依次检验 - 使用原始列名
p1, v1 = chi2_effect(data['age_group'], data['hospital_outcome_1alive_0dead'], '
p2, v2 = chi2_effect(data['sex_0male_1female'], data['hospital_outcome_1alive_0d
p3, v3 = chi2_effect(data['episode_number'], data['hospital_outcome_1alive_0dead
```

```
年龄分组 vs outcome: χ²=287.844, p=0.0000, Cramér's V=0.123
性别 vs outcome: χ²=4.352, p=0.0370, Cramér's V=0.015
发作次数 vs outcome: χ²=10.982, p=0.0268, Cramér's V=0.024
```

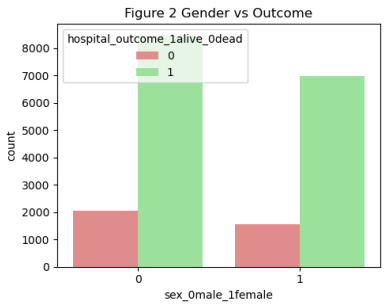
3.可视化：年龄分布与结局

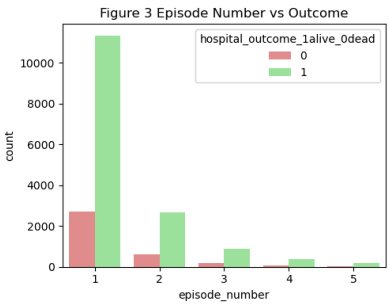
In [14]:
```
plt.figure(figsize=(6,4))
sns.boxplot(x='hospital_outcome_1alive_0dead', y='age_years', data=data,
            palette={'1':'lightgreen', '0':'lightcoral'})
plt.title('Figure 1 Age Distribution by Outcome')
plt.xlabel('Outcome (1=Alive, 0=Deceased)')
plt.ylabel('Age (years)')
plt.tight_layout()
plt.show()
```

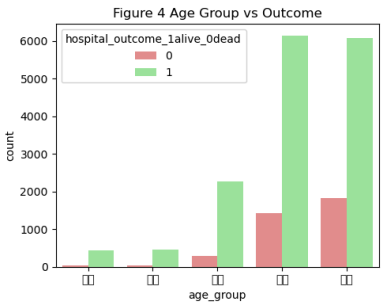


4.可视化：分类变量与结局

In [15]:
```
fig, axes = plt.subplots(1,3, figsize=(15,4))

# 性别
sns.countplot(x='sex_0male_1female', hue='hospital_outcome_1alive_0dead', data=d
              palette={1:'lightgreen',0:'lightcoral'})
axes[0].set_title('Figure 2 Gender vs Outcome')

# 发作次数
sns.countplot(x='episode_number', hue='hospital_outcome_1alive_0dead', data=data
              palette={1:'lightgreen',0:'lightcoral'})
axes[1].set_title('Figure 3 Episode Number vs Outcome')

# 年龄分组
sns.countplot(x='age_group', hue='hospital_outcome_1alive_0dead', data=data, ax=
              palette={1:'lightgreen',0:'lightcoral'})
axes[2].set_title('Figure 4 Age Group vs Outcome')

plt.tight_layout()
plt.show()
```







5.结论汇总

In [16]:
```
alpha = 0.05
conclusions = {
    '年龄分组': '显著' if p1 < alpha else '不显著',
    '性别': '显著' if p2 < alpha else '不显著',
    '发作次数': '显著' if p3 < alpha else '不显著'
}
print('假设检验结论（α=0.05）')
for k,v in conclusions.items():
    print(f'{k}与住院结局关联性：{v}')
```

```
假设检验结论（α=0.05）
年龄分组与住院结局关联性：显著
性别与住院结局关联性：显著
发作次数与住院结局关联性：显著
```